Valuation Models

Richared P. Rumelt

The direct approach to valuation is to forecast the future expected levels of free cash flow (FCF) and then to discount these values at the weighted average cost of capital. However, in many cases we prefer to work with a model in which the choice of a few parameters implies the whole future pattern of cash flows. We shall describe several examples, ranging from the simple to the complex.

The Perpetuity Model

The simplest valuation model is the perpetuity model where NOPLAT is expected to remain constant forever. Since there is no growth, there is no net reinvestment in the firm, so NOPLAT is the (potential) free cash flow and, hence, the dividend payment, were the business a freestanding firm. In this case the value of the firm is

$$V = \frac{P_1}{k} \tag{1}$$

where P_1 is the value of NOPLAT for the coming year and every year thereafter. Here, because there is no growth, the instantaneous and period values are identical.

The Constant Growth Model

Suppose the firm's NOPLAT is expected to grow at a constant instantaneous rate g and that the rate of return on ***new*** incremental investment is r [note this is not the same as asserting that the return on all invested capital, old and new, is *r*). Suppose that at time $t = 0$ we are looking forward and project that the value of NOPLAT over the coming 12 months will be P_1. Additionally, a portion b of NOPLAT is retained and re-invested in the firm. The following year the return *r* on the new investment $b P_1$ raises NOPLAT to the new level $P_2 = P_1(1 + rb)$. Because growth is constant at rate g it must also be that $P_2 = P_1(1 + g)$. Thus we have identity

$$g = rb \tag{2}$$

Hence, the free cash flow F_t can be written in terms of growth and return rather than return and retention:

$$F_t = P_t(1 - b) = P_t(1 - g / r) \tag{3}$$

If the firm grows forever at rate *g,* where $g < k$, the value of the firm is given by

$$V = \frac{P_1(1 - g / r)}{k - g}, \qquad g < k \tag{4}$$

©2010 Richard P. Rumelt. This work is licensed under a Creative Commons Attribution Non-Commercial 4.0 International License (CC BY-NC 4.0). You may copy, distribute, and adapt this material for non-commercial purposes, provided appropriate credit is given. Commercial use requires written permission from the author.

(The subscript 1 on P indicates that the appropriate value is that of the coming year.) Of course if growth is zero, this equation reduces to the perpetuity equation. More interesting is the fact that if $r = k$, then the formula also reduces to the perpetuity equation above (try it). This means that if the rate of return equals the cost of capital, the rate of growth is immaterial.

Note that V is the value of the entire business: stock value plus debt value plus the value of preferred stock and bank loans. To obtain the equity value of the firm it is necessary to deduct the value of the outstanding debt B:

$$S = V - B \tag{5}$$

Finally, note that this valuation is the operating value of the business. In computing NOPLAT and invested capital, we have taken out excess cash and the income on excess cash. To obtain a prediction of the market value of the firm's equity it is necessary to add back these non-operating assets:

$$\begin{aligned} V_M &= V + \text{ExCash} \\ S_M &= V_M - B \end{aligned} \tag{6}$$

The Sales Margin Model

Sometimes it is more intuitive to work with the net sales margin m than the rate of return on investment r. In this case we need to note that

$$r = m / J \tag{7}$$

where J is the marginal ratio of investment to sales (the amount that must be invested to support an additional unit of sales). Given this relationship between r and m, and constant growth in sales, the free cash flow at time t can be written as

$$\begin{aligned} F_t &= P_t(1 - g / r) \\ &= mS_t(1 - gJ / m) \\ &= S_t(m - gJ) \end{aligned} \tag{8}$$

Discounting gives

$$V = \frac{S_1(m - gJ)}{k - g} \tag{9}$$

The Declining Margin Model

From (7) we can define the "normal long-term" sales margin as $m_\infty = kJ$, the sales margin corresponding to the cost of capital. An intuitively appealing model of the sales margin would have its current "abnormal" level moving over time to become the "normal" level. To obtain this behavior, let the sales margin m_t in year t be

$$m_t = m_\infty + \frac{m_\Delta}{(1 + w)^{t-1}} \tag{10}$$

where w is the rate of decay of the abnormal additional margin m_Δ (note that m_Δ can be negative). In presenting the result of this assumption, it is useful to define a "modified" growth rate

$$g' = \frac{1+g}{1+w} - 1. \tag{11}$$

Now, the free cash flow in year t can be written as

$$F_t = S_1(1+g)^{t-1}\left[m_\infty + \frac{m_\Delta}{(1+w)^{t-1}} - gJ\right] \tag{12}$$

This expression looks imposing, but it can be broken into two parts.

$$F_t = (m_\infty - gJ)S_1(1+g)^{t-1} + m_\Delta S_1(1+g')^{t-1}$$

Each part is a fixed quantity growing at a constant (though different) rate. We know (ASA Technical Note on PV Formula) that the present value of $1 growing at rate g is 1/(k-g). Hence,

$$V = \frac{(m_\infty - gJ)S}{k-g} + \frac{m_\Delta S}{k-g'} \tag{13}$$

The first portion of (13) is the value due to the long-term margin and long-term growth rate. The second term is the extra value due to the excess, though declining, component of margin.

Two-Stage Growth Model

The problem with the constant growth model, or the declining margin model, is that the rate of growth must be moderate. If $g > k$ in the constant growth model, or if $(1+g) > (1+k)(1+w)$ in the declining margin model, they "blow up" and cannot be used for valuation. An alternative valuation formula may be derived from the assumption that one level of growth and return continues for N years after which growth and return change to other levels. In particular, this structure allows one to model high growth situations (involving negative free cash flows) as long as they eventually end and are followed by lower growth situations.

Let g_S and r_S denote the "supernormal" levels of growth and return and let g and r denote the "normal" levels. First define

$$H = \left(\frac{1+g_s}{1+k}\right) \tag{14}$$

then the value of the firm is:[1]

$$V = P_1\left[\frac{(1-g_s/r_s)}{k-g_s}\left(1-H^N\right) + H^N\frac{(1-g/r)}{k-g}\right], \tag{15}$$

In applying this model it is important not to let r or g be "too" large or set r to be too small. That is, r should be close to k and g should not be much more than GNP growth. One common assumption is that $g = 0$, or equivalently, that $r = k$. With this simplification, the model becomes

[1]The version of this formula provided by Copeland, Koller, and Murrin (page 278) is incorrect. The exponents in their version should be changed from *N-1* to *N*.

$$V_{k=r} = P_1\left[\frac{(1-g_s/r_s)}{k-g_s}\left(1-H^N\right)+\frac{H^N}{k}\right] \tag{16}$$

In both (15) and (16) the formulae will "blow up" if $k = g_s$. However, this is an artifact of how the present value is expressed. For the case $k = g_s$, the values are

$$V_{k=g_s} = P_1\left[N\frac{\left(1-g_s/r_s\right)}{1+k}+H^N\frac{(1-g/r)}{k-g}\right], \tag{17}$$

and

$$V_{k=g_s;k=r} = P_1\left[N\frac{\left(1-g_s/r_s\right)}{1+k}+\frac{H^N}{k}\right]. \tag{18}$$

Two Stage Sales-Based Model

By using the relationship expressed in (7) that $r = m/J$, the two stage growth model can also be expressed in terms of sales growth and abnormal levels of margin *m* and the incremental capital-to-sales ratio *J*:

$$\begin{aligned} V &= S_1\left[\frac{(m_s-g_sJ_s)}{k-g_s}\left(1-H^N\right)+H^N\frac{(m-gJ)}{k-g}\right] \\ &= S_1m_s\left[\frac{(1-g_sJ_s/m_s)}{k-g_s}\left(1-H^N\right)+H^N\frac{m}{m_s}\frac{(1-gJ/m)}{k-g}\right] \end{aligned} \tag{19}$$

In the first form of (19), the expression in brackets is the value-to-sales ratio. In the second firm, the expression in brackets is the value-to-Noplat ratio, the inverse of the unlevered PE.

Here, the analyst must estimate or specify the long-term Noplat/Sales ratio *m*, the long-term growth rate *g*, and the long incremental capital-to-sales ratio *J*, keeping in mind that the return on incremental invested capital is $r = m/J$. The analyst must than estimate or specify short-term levels of these three parameters, m_s, g_s, and J_s, keeping in mind that $r_s = m_s/J_s$. Finally, the analyst must specify the number of years *N* over which the short-term conditions hold.

Estimating J and r

The traditional and standard way of estimating ROIC is the ratio of NOPLAT to InvestedCapital. This ratio is best calculated based on a one-year lagged value of InvestedCapital, or an average level of InvestedCapital over the past year. The idea is simply that today's earnings are the fruits of yesterday's capital.

One problem with this approach is that the valuation models want the incremental return on new investment, not the ratio of all return to all past investment. Sticking with the gross simplification that investment pays off soon after it is made, it is helpful to look at the estimation of *J* in the margin-based valuation models. *J* is the incremental investment required to support an extra $1 in

sales. In situations where this is the better model, we think of investment as being induced by growth in sales, rather than driving growth. One way of estimating J is to realize that

$$J = \frac{\Delta \text{InvCap}}{\Delta \text{Sales}} = \frac{\Delta C}{\Delta S} \tag{20}$$

which can be manipulated to become

$$J = \frac{\Delta C / C}{\Delta S / S} \frac{C}{S} = J_{\text{avg}} \frac{G_C}{G_s}. \tag{21}$$

Thus, the incremental capital-to-sales ratio is the average *level* of capital-to-sales times the ratio of the growth rate of capital to the growth rate of sales. Each of these can be estimated from several years of data.